

Region Resonances

Data Appendix

Source Data, Tables, and Reference Materials

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Companion documents: Summary · Technical Appendix · Methods Primer · Data Appendix · Region Reference (Identity)

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Table of Contents

- D.1 Region Roster (59 regions, with metaphors)
- D.2 Embedding Model Specification
- D.3 Calibration Test Set and Operating Range
- D.4 Display Transformation Constants
- D.5 Distribution of Top- k Cosine Similarities

D.1 Region Roster

The 59 regions analysed by Region Resonances, listed with their canonical soul metaphors. The corpus is identical to that used by Region Affinities; the metaphors are drawn from the canonical Layer 1 identity descriptions and serve as the first-token anchor of each region's embedded passage.

#	Region	Country	Metaphor
1	Kamptal	Austria	Discipline
2	Steiermark	Austria	Clarity
3	Wachau	Austria	Monumentality
4	Wagram	Austria	Earth
5	Dalmatian Coast	Croatia	Tranquility
6	Alsace	France	Duality
7	Beaujolais	France	Joy
8	Bordeaux	France	Business
9	Burgundy	France	Devotion
10	Champagne	France	Society
11	Châteauneuf-du-Pape	France	Family
12	Jura	France	Eccentricity
13	Loire	France	Sentimentality
14	Northern Rhône	France	Solitude
15	Provence	France	Pleasure
16	Baden	Germany	Warmth
17	Mosel	Germany	Poetry
18	Nahe	Germany	Subtlety
19	Pfalz	Germany	Generosity
20	Rheingau	Germany	Nobility
21	Macedonia	Greece	Austerity
22	Santorini	Greece	Survival
23	Tokaj	Hungary	Melancholy
24	Alto Adige	Italy	Precision
25	Campania	Italy	Memory
26	Etna	Italy	Awakening
27	Friuli-Venezia Giulia	Italy	Dialogue
28	Liguria	Italy	Intimacy
29	Piedmont	Italy	Philosophy
30	Sardinia	Italy	Stubbornness
31	Sicily	Italy	Resurrection
32	Tuscany	Italy	Art
33	Veneto	Italy	Commerce
34	Douro	Portugal	Endurance
35	Goriška Brda	Slovenia	Fortune
36	Catalonia	Spain	Identity
37	Galicia	Spain	Longing (Morriña)
38	Ribera del Duero	Spain	Severity
39	Rioja	Spain	Patience
40	Mendoza	Argentina	Reinvention
41	Patagonia	Argentina	Extremity
42	Barossa Valley	Australia	Fortitude
43	Hunter Valley	Australia	Defiance
44	Margaret River	Australia	Composure
45	Maipo Valley	Chile	Pride
46	Central Otago	NZ	Adventure
47	Hawke's Bay	NZ	Confidence
48	Marlborough	NZ	Assertion
49	Stellenbosch	S. Africa	Aspiration
50	Swartland	S. Africa	Rebellion
51	Columbia Valley	USA	Determination
52	Finger Lakes	USA	Conviction
53	Napa Valley	USA	Ambition
54	Paso Robles	USA	Independence
55	Santa Barbara	USA	Serendipity
56	Santa Cruz Mountains	USA	Obsession
57	Sonoma	USA	Authenticity
58	Walla Walla	USA	Community
59	Willamette Valley	USA	Idealism

D.2 Embedding Model Specification

The embedding model used throughout is OpenAI’s `text-embedding-3-small`. Specifications relevant to the Region Resonances deployment:

Property	Value
Model identifier	<code>text-embedding-3-small</code>
Output dimension	$d = 1536$
Output length	Unit-normalised; $\ E(s)\ _2 = 1$ for every input s
Input language	English (multilingual capability not used for matching)
Maximum input tokens	8,191 (well above any region passage length)
Determinism	Same input always produces the same output
Treatment in pipeline	Fixed, opaque function; weights and architecture not introspected

D.3 Calibration Test Set and Operating Range

The display transformation constants $s_{\min}$ and $s_{\max}$ were calibrated against a test set of 60 queries spanning four query types.

Query type	n	Examples
Single abstract noun	18	resilience, devotion, melancholy, pleasure
Compound abstract phrase	16	old soul, quiet ambition, inherited burden
Sensory / atmospheric phrase	14	the feeling of late summer, dust on stone
Short prose fragment	12	wines that ask you to wait
Total	60	—

For each query, the full 59-element similarity vector was computed and the empirical distribution of cosine values recorded. Across all 60 queries, observed similarities fell almost entirely within $[0.15, 0.65]$. Distribution summary:

Statistic	Value
Min observed cosine (across all queries, all regions)	0.08
1st percentile	0.16
Mean across all (query, region) pairs	0.41
99th percentile	0.66
Max observed cosine (excluding paraphrase queries)	0.78
Max observed cosine (paraphrase queries, $n = 4$)	0.91

Pathological inputs (random byte sequences, single-character queries, queries in languages other than English) routinely produce cosine values below 0.15. These are clipped to the lower display boundary by the transformation in Section D.4.

D.4 Display Transformation Constants

The clip-and-rescale transformation is parameterised by two constants:

$$\tilde{s}_i = \text{clip}\left(\frac{s_i - s_{\min}}{s_{\max} - s_{\min}}, 0, 1\right) \times 100.$$

Constant	Value
$s_{\min}$ (lower clip boundary)	0.15
$s_{\max}$ (upper clip boundary)	0.70
Display range (clipping aside)	[0, 100]
Cosine difference per display unit	0.0055
Display value for chance similarity ($s = 0.30$)	27
Display value for typical strong match ($s = 0.55$)	73

The $s_{\min}$ value is set just above the chance baseline observed for unrelated queries; values below this are clipped to 0 to signal that the query has fallen outside the matcher’s effective scope. The $s_{\max}$ value is set so that paraphrase queries (which produce cosine values above 0.70) saturate the display at 100; this prevents users from inferring fine distinctions among very-strong matches that the underlying cosine cannot reliably discriminate.

D.5 Distribution of Top- k Cosine Similarities

For the calibration test set, the distribution of top-1 and top-5 cosine similarities is summarised below. The top-1 distribution is the cosine of the best match for each query; the top-5 distribution is the cosine of the fifth-best match.

Statistic	Top-1	Top-5
Min	0.42	0.32
25th %ile	0.51	0.41
Median	0.58	0.46
Mean	0.59	0.47
75th %ile	0.65	0.52
Max	0.78	0.60

The typical top-1 to top-5 spread for a content-bearing query is approximately 0.10 in cosine units — wide enough that the top match is consistently distinguishable from the fifth, while the fifth is consistently distinguishable from the next-best (sixth) region. After display transformation, top-1 scores range from approximately 50 to 100 with a median of 78; top-5 scores range from 30 to 80 with a median of 56. The display transformation preserves this rank-discriminating structure exactly.

References

- OpenAI (2024). New embedding models and API updates. Technical announcement covering `text-embedding-3-small` and `text-embedding-3-large`.

- Reimers, N. & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, 3982–3992.